REPORT TITLE

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STUDENT'S NAME
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INDUX

INTRODUCTION

This report presents a digital forensic investigation conducted on two forensic images, **E_01_Physical_Image_Christy** and **USB_03.E01**. The objective of this investigation is to identify, recover, and analyse digital artefacts related to suspected piracy and other potential criminal activities.

The investigation was carried out using two virtual machine environments: **Kali Linux** and **Windows 11 Home**. A range of industry-standard digital forensic tools were utilised, including Autopsy Forensic Browser, FTK Imager, VeraCrypt, and HxD. These tools enabled the systematic examination, recovery, and analysis of digital evidence throughout the investigation.

Tools Used

- **FTK Imager** – Used for imaging, mounting and recovering deleted files.
- **Autopsy** – Used for file analysis, metadata extraction, and timeline investigation.
- **HxD** – Used for hex-level file signature analysis.
- **VeraCrypt** – Used to mount and analyse encrypted containers.

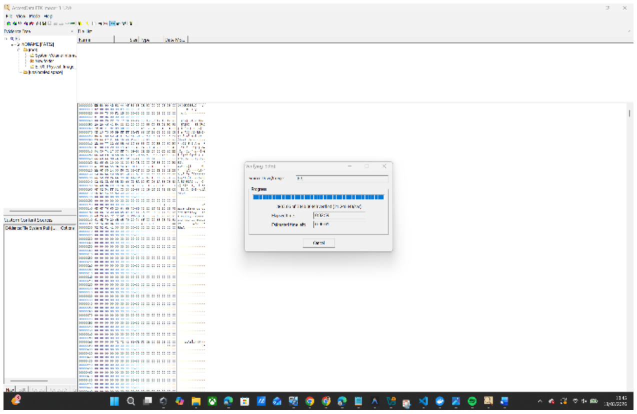
Kali Linux Tools

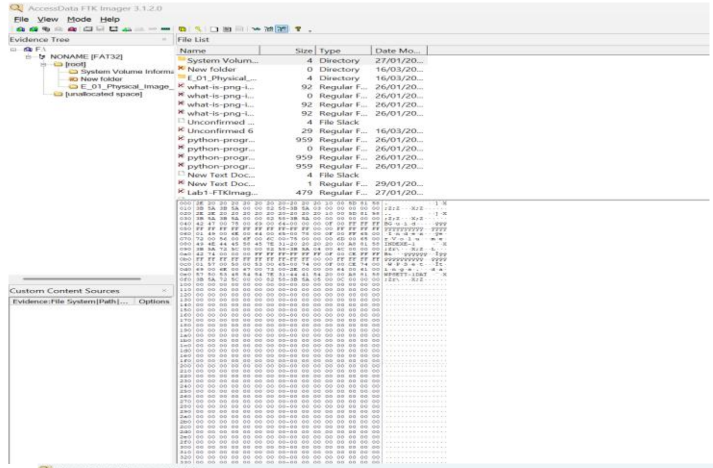
- **ewfmount** – Mount E01 forensic images in Kali Linux
- **mount** – Access file systems in read-only mode
- **sha256sum** – Generate SHA256 hash values
- **xxd** – Display file content in hexadecimal format
- **mediaInfo** – Extract media properties (bit rate, format)
- **FFmpeg** – Analyse and verify media file encoding
- **strings** – Extract readable text from files
- **oremost** – Recover deleted files using file signatures
- **binwalk** – Detect hidden or embedded data
- **Volatility** – Analyse memory (RAM) for hidden processes

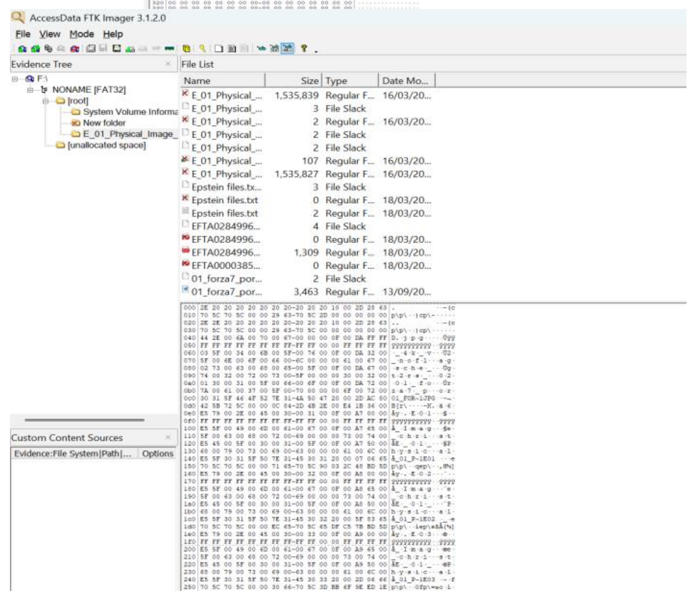
Task 1: Recovering Files from a Forensic Image

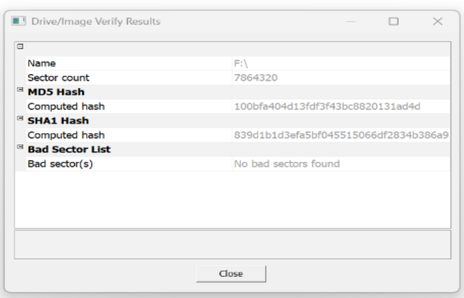
A forensic image of a USB device was examined using **FTK Imager** to recover deleted image files and extract key metadata for analysis.
The contents of the image were inspected, and previously deleted files were successfully identified and exported. Each recovered file was analysed to determine its original file name, file signature, file extension, and file size.
The extracted data was systematically recorded and organised into a table to support documentation and further examination. This task illustrates the application of forensic imaging tools in recovering deleted digital artefacts and preserving evidentiary details.

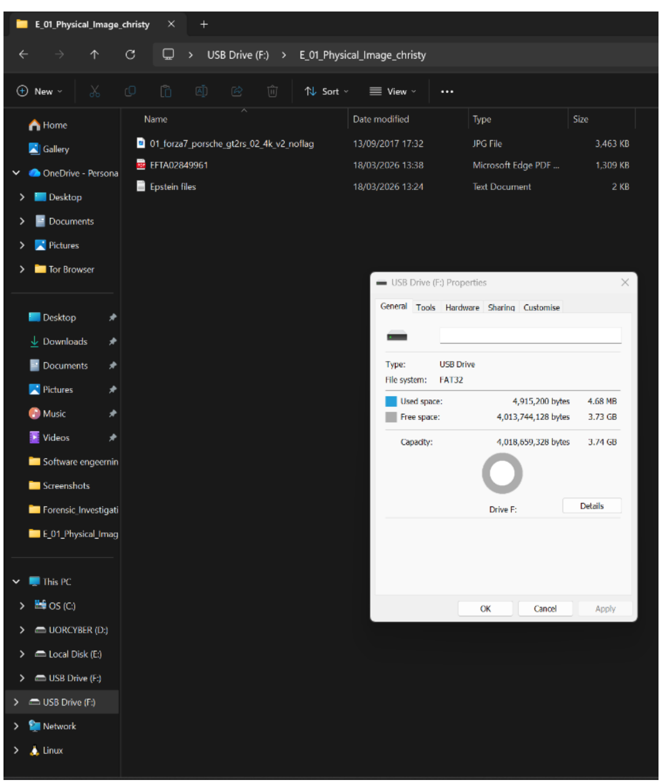
File Name	File Signature (Hex)	File Type/ Extension	File Size
!emdump.mem	52 49 46 C4	.avi / .wav (possible)	508,000 bytes (496 KB)
Python-programming-3840x2160.jpg.crdownload	A2 C1 A6 9B	Unknown / Fragment	959 bytes (0.94 KB)
Efta02731485.pdf	25 50 44 46	.pdf	29 bytes (0.03 KB)
Epstein files.txt	45 70 73 74	.txt	3 bytes (0.003 KB)
filters.format	52 49 46 46	RIFF (.avi / .wav possible)	119 bytes (0.12 KB)
01_forza7_porsche_gt2rs_02_4k_v2_noflag.jpg	FF D8 FF E1	.jpg	3,463 bytes (3.38 KB)
E_01_Physical_Image_christy.E02	80 16 28 D8	Unknown (possible corrupted file	1,535,839bytes (1.46 MB)
Unconfirmed 6	25 50 44 46	.pdf	29 bytes (0.03KB)











USB Capacity: **3.73 GB**

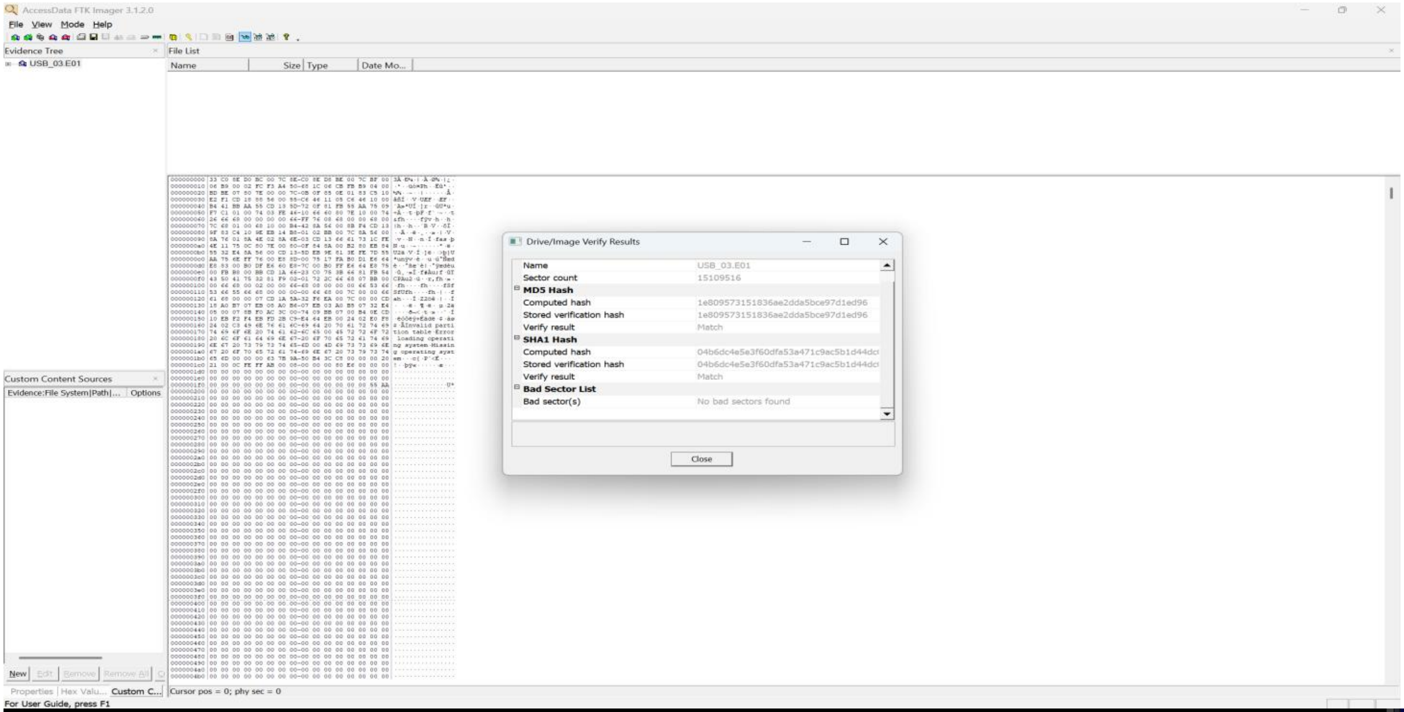
File saved as: **E_01_Physical_Image_Christy**

Task2: File Signature Investigations – Analysing the USB_03_E01 image

This task involved analysing the forensic image **USB_03.E01** to identify file types through file signature investigation. The analysis was conducted in both **Kali Linux** and **Windows** environments using tools such as **FTK Imager** and **HxD** to examine files at the hexadecimal level.

The goal was to compare the performance, usability, and accuracy of each platform in determining true file formats. This approach helped identify the most efficient method for digital evidence analysis.

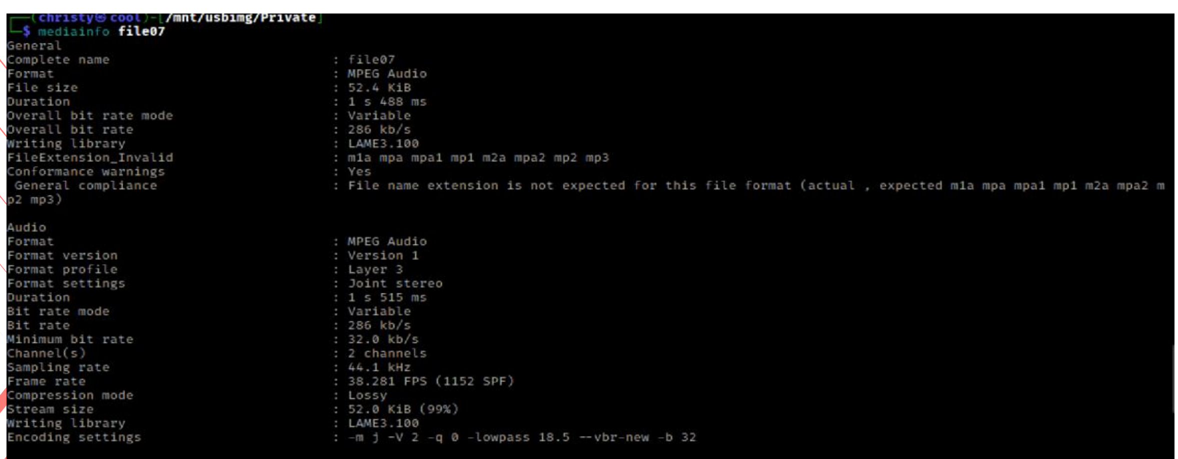
2.1 The forensic image USB_03.E01 was loaded into FTK Imager and successfully verified.



2.2

File Name	File Signature	File Extension	File Sizes
file - 1	50 4B	.zip	13 KB
file - 2	25 50 44 46	.pdf	464 KB
file - 01	25 50 44 46	.pdf	126 KB
file - 02	50 4B	.zip	13,214 KB
file - 03	74 65 78 74	.txt	3 KB
file - 04	3C 68 74 6D	.html	79 KB
file - 05	37 7A BC AF	.7z	1,643 KB
file - 06	47 49 46	.gif	1,688 KB
file - 07	FF FB	.mp3	53 KB

Kali Linux: **mediainfo** tool used to get the details from **file07.**



```
┌──(christy㉿cool)-[/mnt/usbimg/Private]
└─$ mediainfo file07
General
Complete name                            : file07
Format                                   : MPEG Audio
File size                                : 52.4 KiB
Duration                                 : 1 s 488 ms
Overall bit rate mode                    : Variable
Overall bit rate                         : 286 kb/s
Writing library                          : LAME3.100
FileExtension_Invalid                    : m1a mpa mpa1 mp1 m2a mpa2 mp2 mp3
Conformance warnings                     : Yes
 General compliance                      : File name extension is not expected for this file format (actual , expected m1a mpa mpa1 mp1 m2a mpa2 m
p2 mp3)

Audio
Format                                   : MPEG Audio
Format version                           : Version 1
Format profile                           : Layer 3
Format settings                          : Joint stereo
Duration                                 : 1 s 515 ms
Bit rate mode                            : Variable
Bit rate                                 : 286 kb/s
Minimum bit rate                         : 32.0 kb/s
Channel(s)                               : 2 channels
Sampling rate                            : 44.1 kHz
Frame rate                               : 38.281 FPS (1152 SPF)
Compression mode                         : Lossy
Stream size                              : 52.0 KiB (99%)
Writing library                          : LAME3.100
Encoding settings                        : -m j -V 2 -q 0 -lowpass 18.5 --vbr-new -b 32
```

```
┌──(christy㉿cool)-[/mnt/usbimg/Private]
└─$ ffmpeg -i file07
ffmpeg version 8.0.1-3+b1 Copyright (c) 2000-2025 the FFmpeg developers
  built with gcc 15 (Debian 15.2.0-13)
  configuration: --prefix=/usr --extra-version=3+b1 --toolchain=hardened --libdir=/usr/lib/x86_64-linux-gnu --incdir=/usr/include/x86_64-linux-gnu
 --arch=amd64 --enable-gpl --disable-stripping --disable-pocketsphinx --disable-libcaca --disable-libmfx --disable-omx --enable-gnutls --enable-li
baom --enable-libass --enable-libbs2b --enable-libcdio --enable-libcodec2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libf
reetype --enable-libfribidi --enable-libglslang --enable-libgme --enable-libgsm --enable-libharfbuzz --enable-libmp3lame --enable-libmysofa --enab
le-libopenjpeg --enable-libopenmpt --enable-libopus --enable-librubberband --enable-libshine --enable-libsnappy --enable-libsoxr --enable-libspeex
 --enable-libtheora --enable-libtwolame --enable-libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libx265 --enable-libxml2
--enable-libxvid --enable-libzimg --enable-openal --enable-opencl --enable-opengl --disable-sndio --enable-libvpl --enable-libdc1394 --enable-libd
rm --enable-libiec61883 --enable-chromaprint --enable-frei0r --enable-ladspa --enable-libbluray --enable-libdvdnav --enable-libdvdread --enable-li
bjack --enable-libjxl --enable-libpulse --enable-librabbitmq --enable-librist --enable-libsrt --enable-libssh --enable-libsvtav1 --enable-libx264
--enable-libzmq --enable-libzvbi --enable-lv2 --enable-sdl2 --enable-libplacebo --enable-librav1e --enable-librsvg --enable-shared
  libavutil      60.  8.100 / 60.  8.100
  libavcodec     62. 11.100 / 62. 11.100
  libavformat    62.  3.100 / 62.  3.100
  libavdevice    62.  1.100 / 62.  1.100
  libavfilter    11.  4.100 / 11.  4.100
  libswscale      9.  1.100 /  9.  1.100
  libswresample   6.  1.100 /  6.  1.100
Input #0, mp3, from 'file07':
  Duration: 00:00:01.45, start: 0.025057, bitrate: 295 kb/s
  Stream #0:0: Audio: mp3 (mp3float), 44100 Hz, stereo, fltp, 288 kb/s, start 0.025057
    Metadata:
      encoder         : LAME3.100
At least one output file must be specified
```

Tool

```
┌──(christy㉿cool)-[/mnt/usbimg/Private]
└─$ echo "obase=2;195" | bc
11000011

┌──(christy㉿cool)-[/mnt/usbimg/Private]
└─$ mediainfo file07
General
Complete name                            : file07
Format                                   : MPEG Audio
File size                                : 52.4 KiB
Duration                                 : 1 s 488 ms
Overall bit rate mode                    : Variable
Overall bit rate                         : 286 kb/s
Writing library                          : LAME3.100
FileExtension_Invalid                    : m1a mpa mpa1 mp1 m2a mpa2 mp2 mp3
Conformance warnings                     : Yes
 General compliance                      : File name extension is not expected for this file format (actual , expected m1a mpa mpa1 mp1 m2a mpa2 m
p2 mp3)

Audio
Format                                   : MPEG Audio
Format version                           : Version 1
Format profile                           : Layer 3
Format settings                          : Joint stereo
Duration                                 : 1 s 515 ms
Bit rate mode                            : Variable
Bit rate                                 : 286 kb/s
Minimum bit rate                         : 32.0 kb/s
Channel(s)                               : 2 channels
Sampling rate                            : 44.1 kHz
Frame rate                               : 38.281 FPS (1152 SPF)
Compression mode                         : Lossy
Stream size                              : 52.0 KiB (99%)
Writing library                          : LAME3.100
Encoding settings                        : -m j -V 2 -q 0 -lowpass 18.5 --vbr-new -b 32
```

2.3: The first three bytes of file07 are FF FB 90. file07 is an MP3 audio file, even if it has no extension.

2.4: Bit rate: 286 kb/s

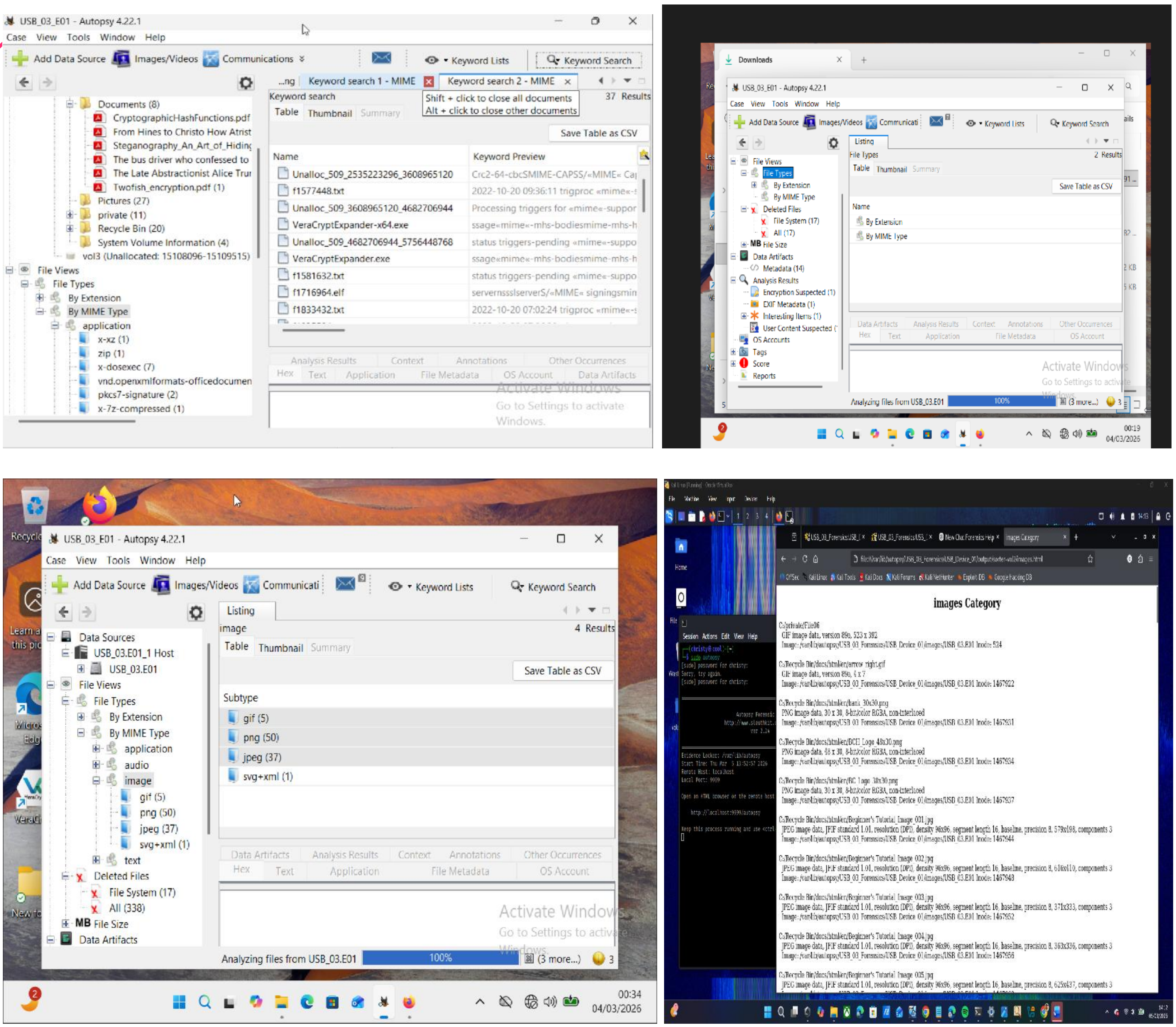
2.5: Convert Bit Rate to Binary: 1000011

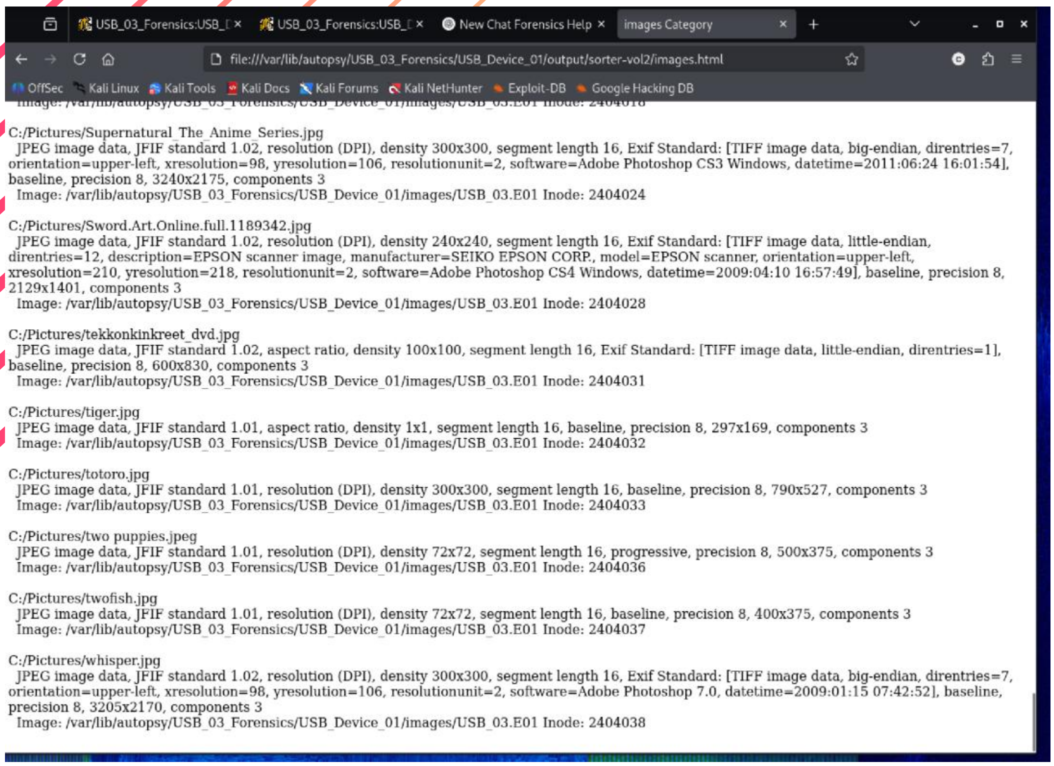
Task 3: Analysis Using Autopsy

Setting up Kali Linux took approximately one hour, after which Autopsy was launched using the **sudo autopsy** command and accessed via **http://localhost:9999/autopsy**. Windows Autopsy proved easier to use by comparison. Both Windows and Kali Linux were operated as virtual machines during the analysis.

3.1: Autopsy case files use the extension: **.aut**

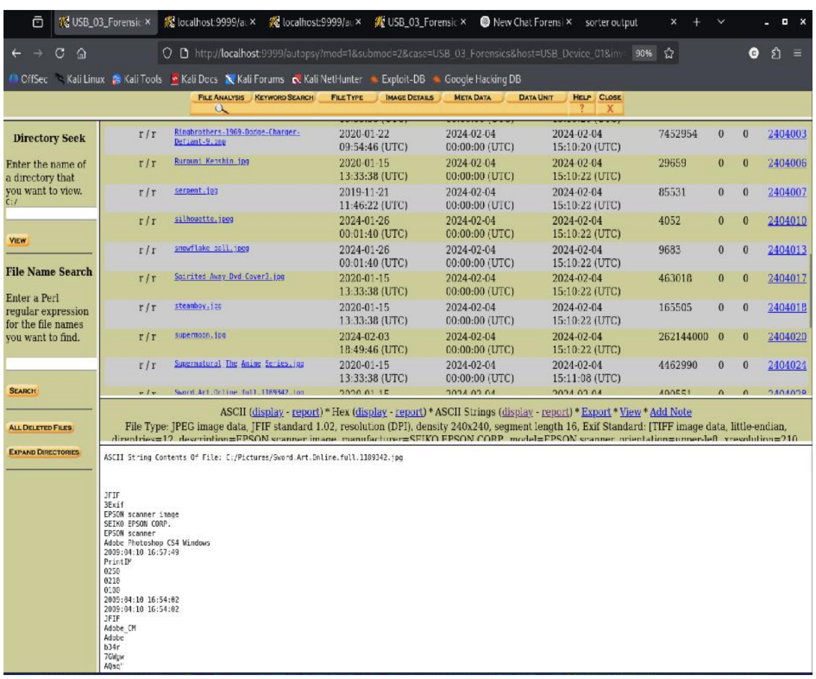
3.2:

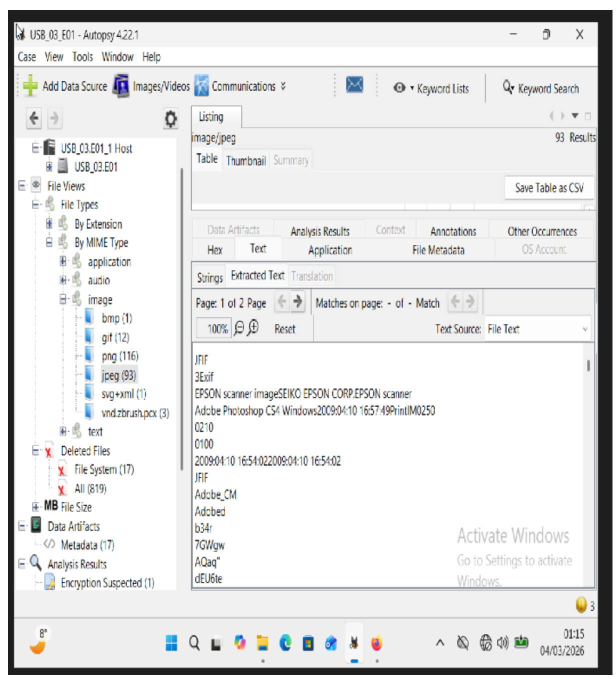




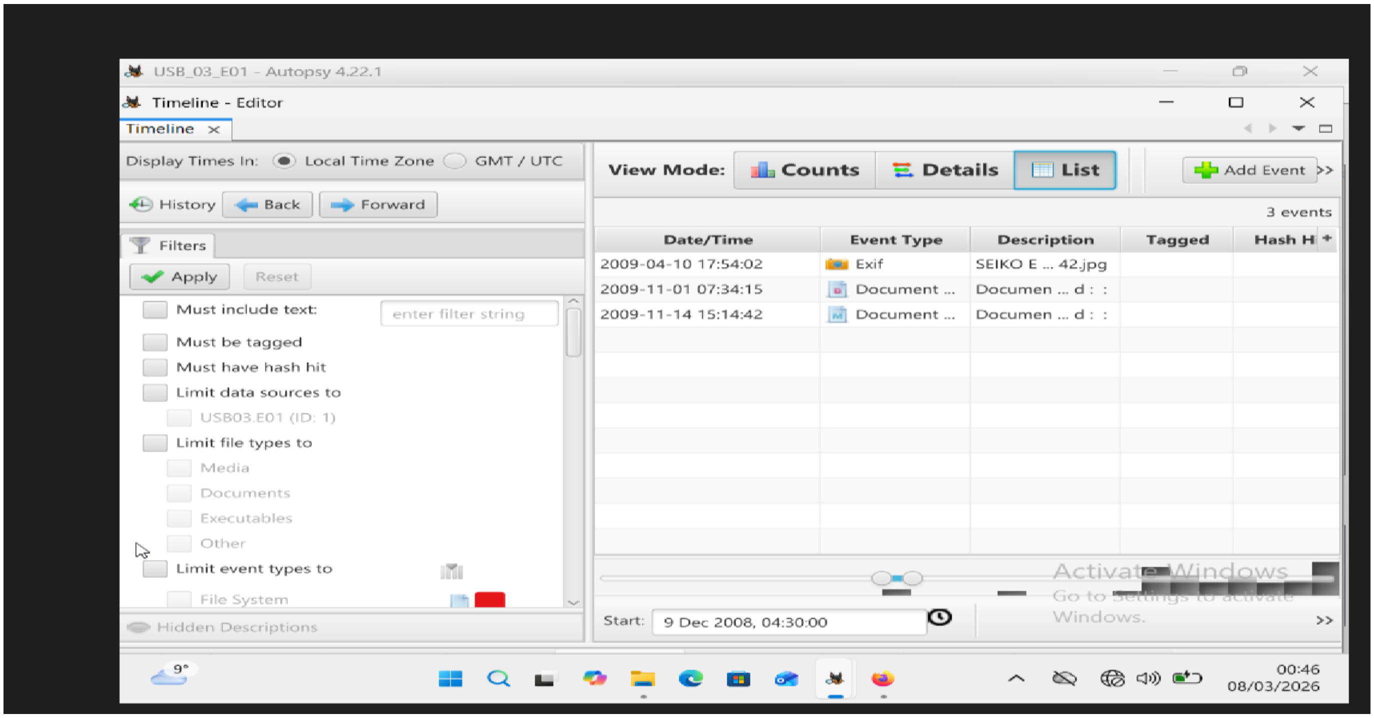
bmp: 1
Jpeg: 10

3.3: EPSON Scanner Image SEIKO.EPSON scanner



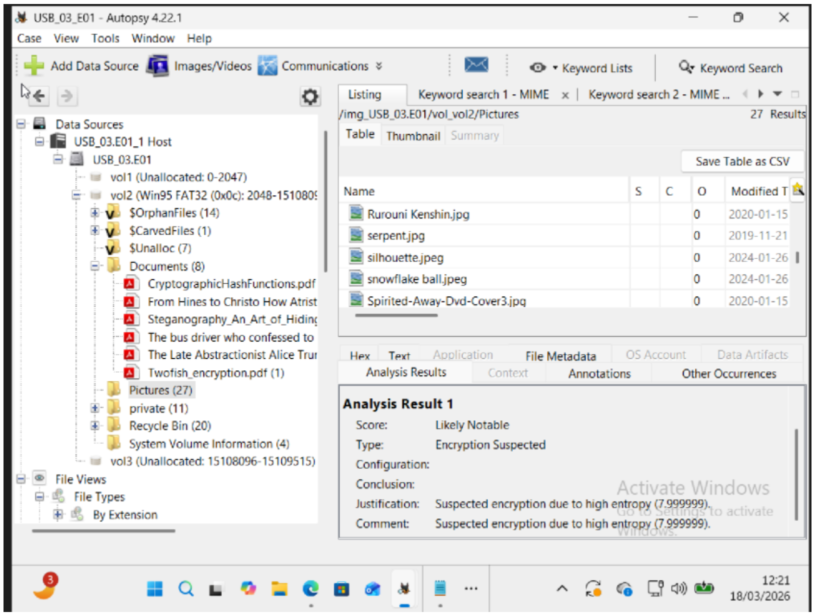


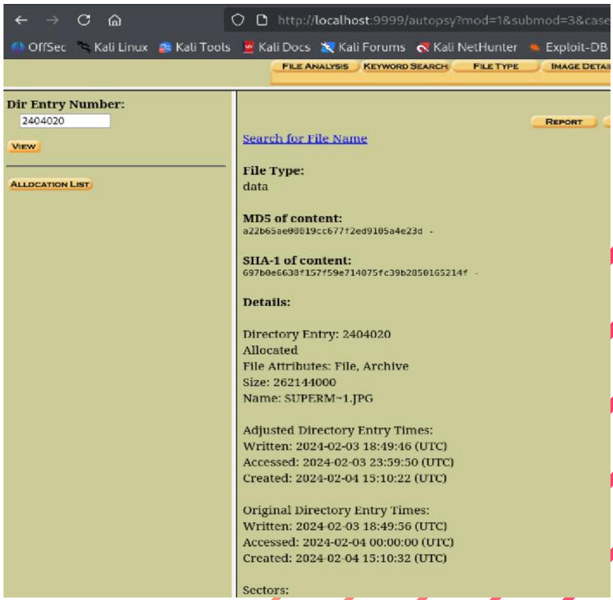
3.4: TimeLine view in 2009 (total 3 events)



- Autopsy ⟶ Tool ⟶ Timeline
- During the analysis using Autopsy on Kali Linux, the Timeline view was found to display events only from **2011 to 2018**, while the **2009 timeline** remained empty.

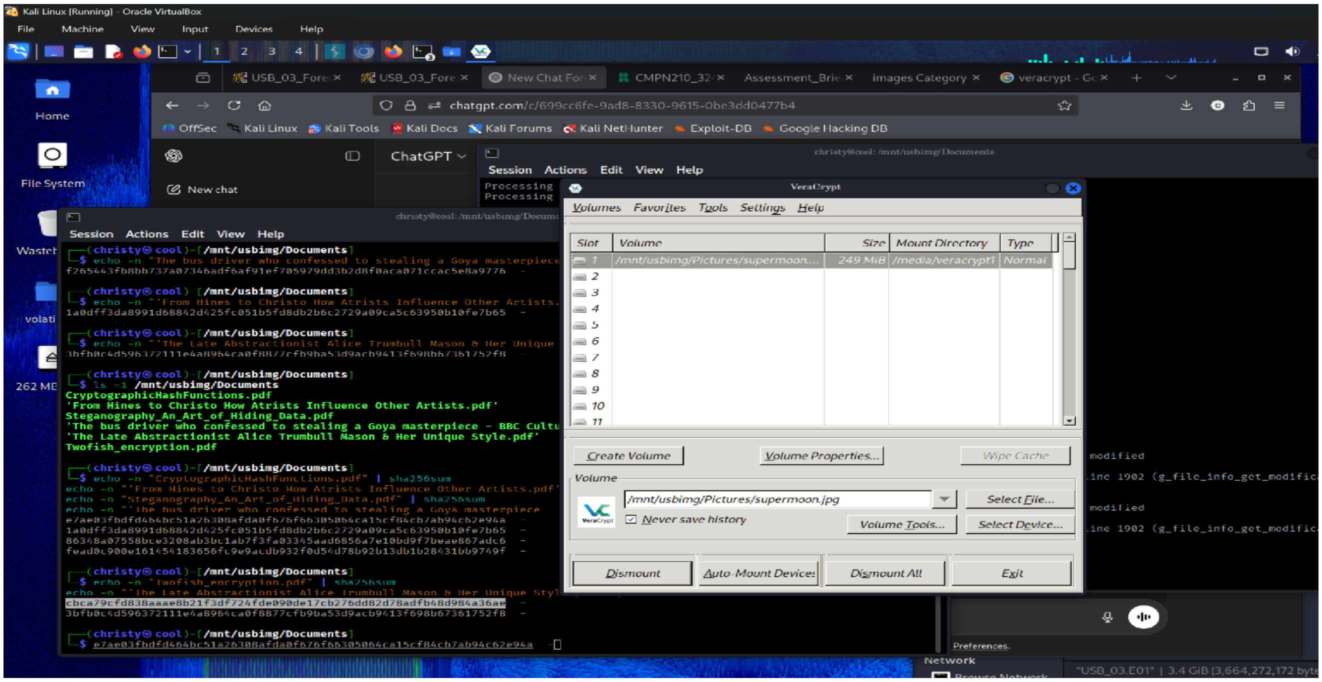
3.5: A suspicious file identified as **supermoon.jpg** was located within the pictures folder. The file exhibited no readable content and appeared to be encrypted.



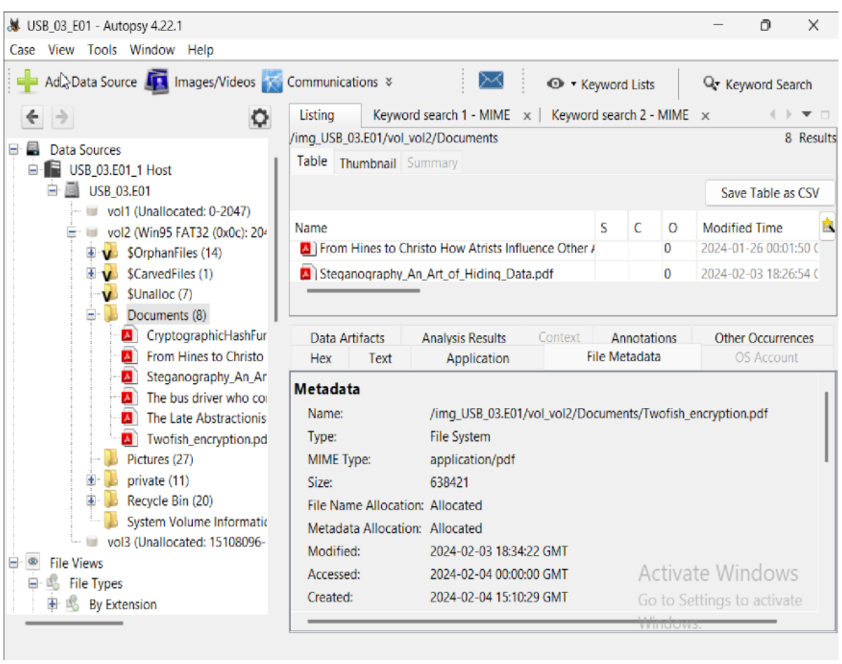


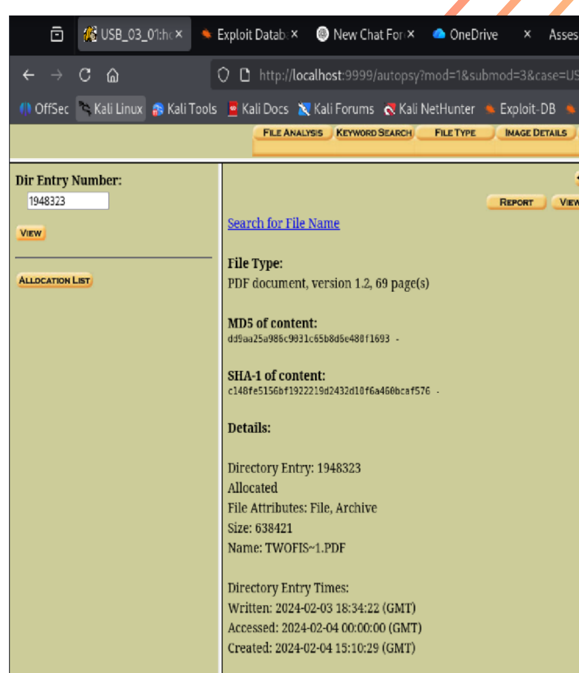
Task4: Encrypted File Analysis

The encrypted file was extracted and mounted using VeraCrypt. Access was achieved using the SHA256 hash cbca79cfd838aaae8b21f3df724fde090de17cb276dd82d78adfb48d984a36ae, which was generated from a file named **Twofish_encryption.pdf** located in the Documents folder.



4.1





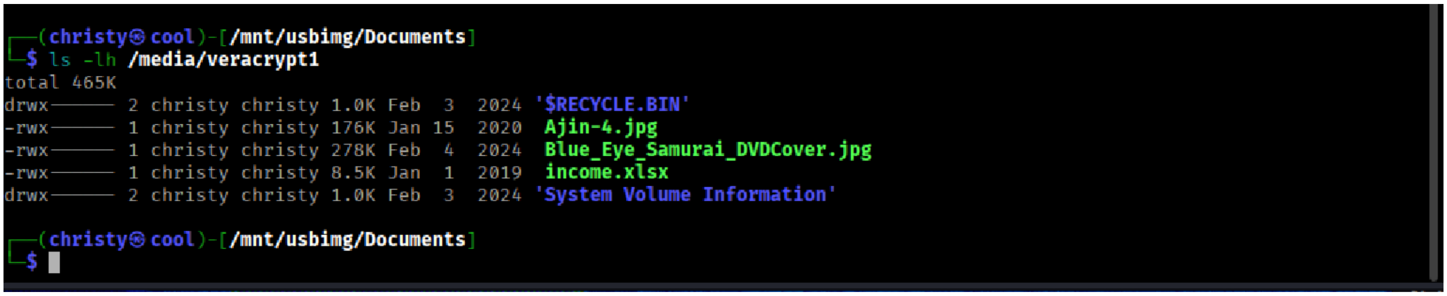
Twofish_encryption.pdf .pdf 638421 bytes

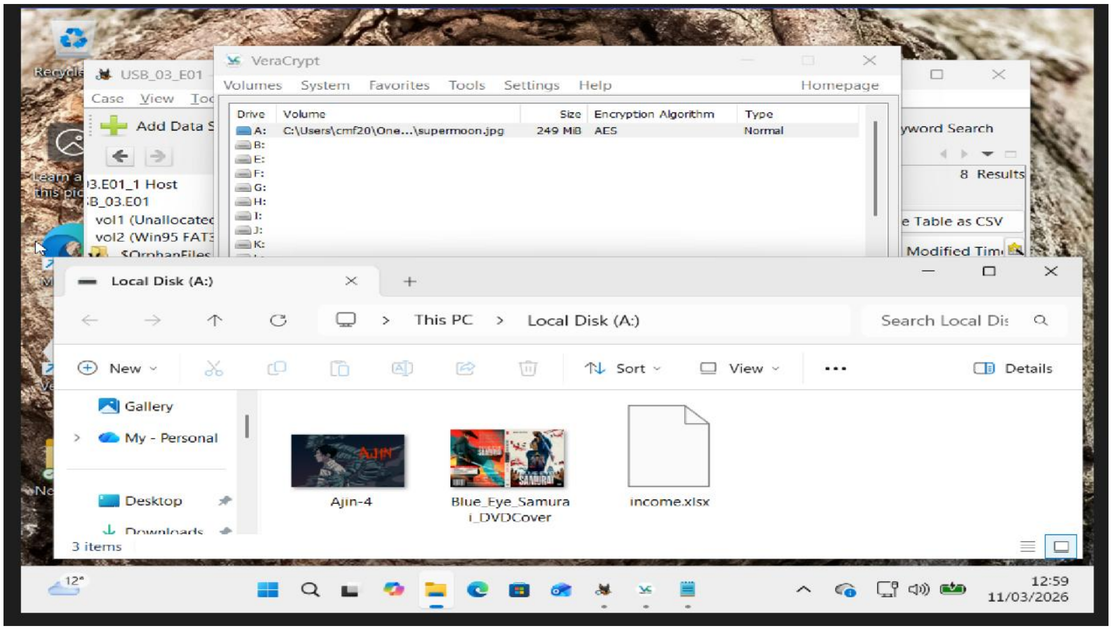
4.2: cbca79cfd838aaae8b21f3df724fde090de17cb276dd82d78adfb48d984a36ae

4.3: Files and Folders in **veracrypt1**

i. Ajin-4.jpg (175.3 KB)
ii. Blue_Eye_Samurai_DVDCover.jpg (277.7 KB)
iii. income.xlsx (8.5 KB)

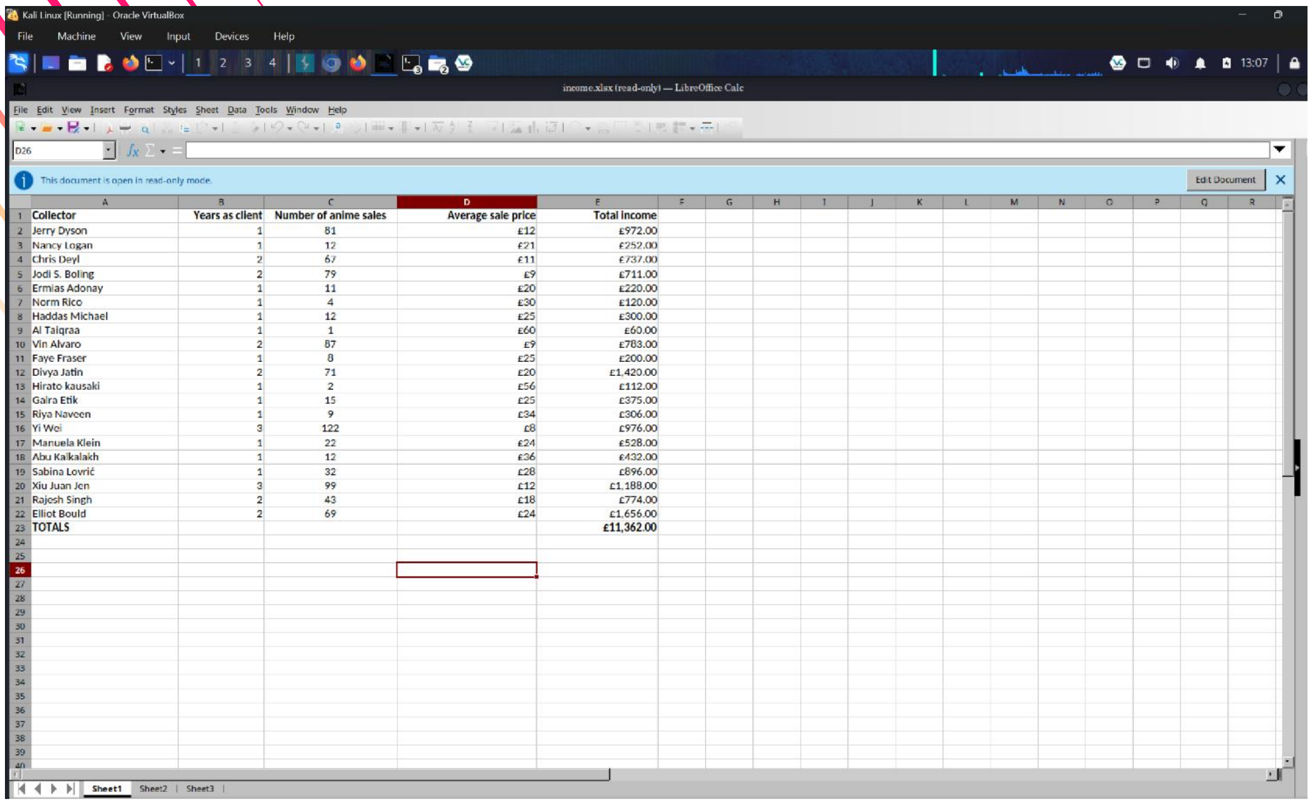
$RECYCLE.BIN (Folder)
i. $IJ5M9ID.jpg (58 bytes)
ii. $IRM3QTJ.jpg (118 bytes)
iii. $RJ5M9ID.jpg (2.6 MB)
iv. $RRM3QTJ.jpg (70.9 KB)
v. $IJKF6BP.xlsx (56 bytes)
vi. $RJKF6BP.xlsx (10.8 KB)





I get access to the RECYCLE.BIN folder for windows from the application **WinRAR-x64**
It was masked inside the file **Blue_Eye_Samurai_DVDCover.jpg.**

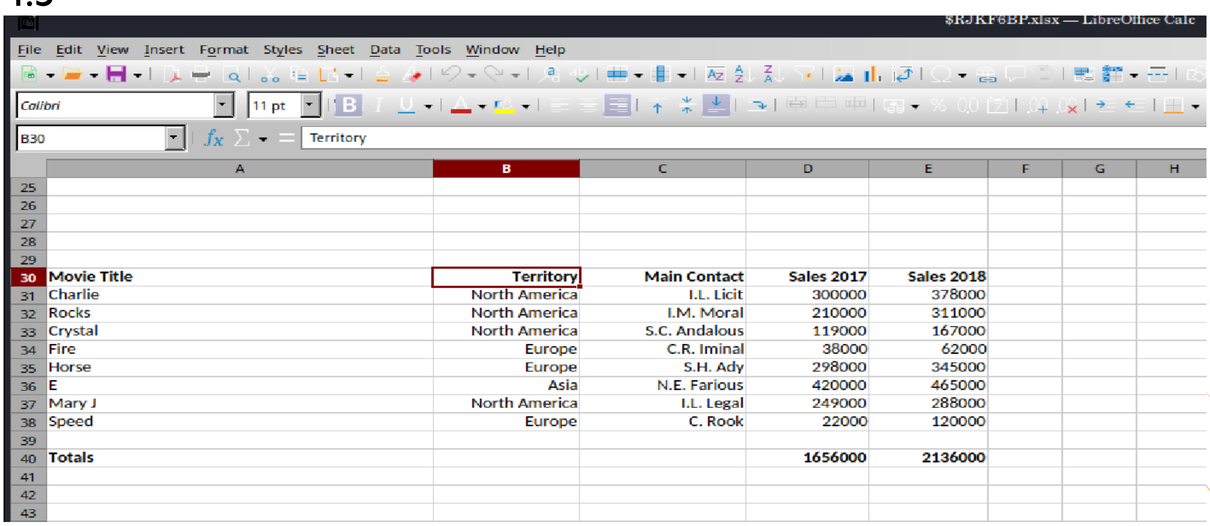
4.4



Collector	Years as client	Number of anime sales	Average sale price	Total income
Jerry Dyson	1	81	£12	£972.00
Nancy Logan	1	12	£21	£252.00
Chris Deyl	2	67	£11	£737.00
Jodi S. Boling	2	79	£9	£711.00
Ermias Adonay	1	11	£20	£220.00
Norm Rico	1	4	£30	£120.00
Haddas Michael	1	12	£25	£300.00
Al Taiqraa	1	1	£60	£60.00
Vin Alvaro	2	87	£9	£783.00
Faye Fraser	1	8	£25	£200.00
Divya Jatin	2	71	£20	£1,420.00
Hirato kausaki	1	2	£56	£112.00
Gaira Etik	1	15	£25	£375.00
Riya Naveen	1	9	£34	£306.00
Yi Wei	3	122	£8	£976.00
Manuela Klein	1	22	£24	£528.00
Abu Kalkalakh	1	12	£36	£432.00
Sabina Lovrić	1	32	£28	£896.00
Xiu Juan Jen	3	99	£12	£1,188.00
Rajesh Singh	2	43	£18	£774.00
Elliot Bould	2	69	£24	£1,656.00
TOTALS				£11,362.00

Total **21collectors** been found in the income.xlsx file.

4.5



Movie Title	Territory	Main Contact	Sales 2017	Sales 2018
Charlie	North America	I.L. Licit	300000	378000
Rocks	North America	I.M. Moral	210000	311000
Crystal	North America	S.C. Andalous	119000	167000
Fire	Europe	C.R. Iminal	38000	62000
Horse	Europe	S.H. Ady	298000	345000
E	Asia	N.E. Farious	420000	465000
Mary J	North America	I.L. Legal	249000	288000
Speed	Europe	C. Rook	22000	120000
Totals			1656000	2136000

There is a total of 8 movie titles listed in the second Excel file. This file is located inside the **$RECYCLE.BIN** folder, has the filename **$RJKF6BP.xlsx** sheet 3, and is 10.8 KB in size.

4.6 **Evidence of Crime**

Based on the forensic analysis of the **USB_03.E01** image and the VeraCrypt container, there is strong evidence of illegal drug-related activity.
The investigation identified files such as **income.xlsx**, which contained financial records, suggesting organised transactions. In addition, coded terms including *"Charlie," "Rocks," "Crystal," "Horse," "E," "Mary J,"* and *"Speed"* were found, which are commonly known street names for controlled drugs.
The presence of anime-related images and media content suggests that anime distribution may have been used as a cover to mask the illegal operation. Furthermore, the use of a VeraCrypt container indicates an attempt to conceal this information.
These findings suggest potential violations of the Misuse of **Drugs Act 1971**. Overall, the evidence indicates involvement in **illegal drug distribution activities disguised under media distribution**.

- **Charlie** → Cocaine
- **Rocks** → Crack cocaine
- **Crystal** → Crystal meth
- **Fire** → High-quality drugs (often cannabis or meth)
- **Horse** → Heroin
- **E** → Ecstasy (MDMA)
- **Mary J** → Marijuana
- **Speed** → Amphetamines

Conclusion

In this investigation, we analysed two digital devices using specialized forensic tools on both Linux and Windows. During the process, we successfully recovered deleted files, analysed file signatures, and extracted useful data.
The breakthrough in the case came when we found a hidden encrypted container. Inside, we discovered detailed financial records and coded language tied to illegal drug distribution—including words like 'Charlie', 'Rocks', 'Crystal', and 'Speed'. We also found that the suspect was using anime media files as a cover for this operation.
Overall, the digital evidence strongly points to the suspect running an organized illegal drug ring disguised behind a media sharing front. This case perfectly demonstrates how essential digital forensics is for uncovering hidden crimes.

References

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- HxD. (n.d.). *HxD Hex Editor Manual*.
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